

HW9 con.

* C51 Method 3:

Row-Reduce:

$$\begin{bmatrix} 2 & 3 & 1 & 5 \\ -1 & 0 & 1 & -1 \\ 2 & 1 & -1 & 3 \end{bmatrix}$$

$$R_1 \leftarrow \frac{1}{2}R_1 : \begin{bmatrix} 1 & 1.5 & 0.5 & 2.5 \\ -1 & 0 & 1 & -1 \\ 2 & 1 & -1 & 3 \end{bmatrix}$$

$$R_2 \leftarrow R_2 + R_1 : \begin{bmatrix} 1 & 1.5 & 0.5 & 2.5 \\ 0 & 1.5 & 1.5 & 1.5 \\ 2 & 1 & -1 & 3 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 2R_1 : \begin{bmatrix} 1 & 1.5 & 0.5 & 2.5 \\ 0 & 1.5 & 1.5 & 1.5 \\ 0 & -2 & -2 & -2 \end{bmatrix}$$

$$R_2 \leftarrow \frac{2}{3}R_2 : \begin{bmatrix} 1 & 1.5 & 0.5 & 2.5 \\ 0 & 1 & 1 & 1 \\ 0 & -2 & -2 & -2 \end{bmatrix}$$

$$R_3 \leftarrow R_3 + 2R_2 : \begin{bmatrix} 1 & 1.5 & 0.5 & 2.5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - \frac{3}{2}R_2 : \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

* Pivots in columns 1 and 2, columns 3 and 4

are linear combinations of them!

linearly independent set T that spans the same space:

$$T = \left\{ \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\langle T \rangle = \langle S \rangle$$